

Kingston
University
London

Kingston
School
of Art

Programme Specification

Title of Course: MA Aesthetics and Art Theory

Date Specification Produced: October 2012

Date Specification Last Revised: August 2018 ~~14.9.17~~

This Programme Specification is designed for prospective students, current students, academic staff and employers. It provides a concise summary of the main features of the programme and the intended learning outcomes that a typical student might reasonably be expected to achieve and demonstrate if they take full advantage of the learning opportunities that are provided. More detailed information on the teaching, learning and assessment methods, learning outcomes and content of each module can be found in the Course Handbook on Canvas and in individual Module Descriptors.

SECTION 1: GENERAL INFORMATION

Title:	<u>MA Aesthetics and Art Theory</u>
Awarding Institution:	<u>Kingston University</u>
Teaching Institution:	<u>Kingston University</u>
Location:	<u>Penrhyn Road/Knights Park</u>
Programme Accredited by:	<u>NA</u>

SECTION2: THE PROGRAMME

A. Programme Introduction

This unique programme combines a grounding in philosophical aesthetics in the modern European tradition with study of contemporary art theory. The programme pays particular attention to the influence of Immanuel Kant's philosophy and aesthetic theory, and to the debates that structured the development of post-Kantian aesthetics in Germany, France and the Anglophone world. In addition to close textual analysis of fundamental works by philosophers and critical theorists, the course involves detailed critical engagement with major developments in modern and contemporary art. The MA appeals to students with a background in fine art, art history or art theory, as well as to students with a background in general or analytical Philosophy, or related subjects like cultural studies, visual culture, and literature. It provides an ideal preparation for doctoral research in art, visual culture or philosophy, and other related fields in the humanities and social sciences. The MA also prepares graduates for a wide range of careers in the arts, education, and the creative industries.

The MA in Aesthetics and Art Theory is taught by internationally recognized specialists at the UK's most dynamic and inventive centre for research in modern European philosophy, the CRMEP. Since its inception in 1994, the CRMEP has developed a national and international reputation for teaching and research in the field of post-Kantian European philosophy, characterised by a strong emphasis on broad cultural and intellectual contexts and a distinctive sense of social and political engagement. Core staff teaching on this programme are recognised as leading philosophers in their fields; unusually, in philosophy, many are also well-known for their writing about contemporary art. The MA programme is enhanced by the CRMEP's lively series of research events, including collaborative projects with UK and European Art Institutions. The CRMEP also has strong links with the Philosophy Department at the University of Paris 8, including staff exchanges and a joint MA. MA Aesthetics and Art Theory students may take advantage of the Erasmus arrangement with Paris 8 and spend part of their time studying there.

The course also includes an optional integrated work placement or placements, which enables students to further develop their professional skills and enhance their employability. From the start of the course, students who select this option will begin to work to secure (a) placement(s) suitable for their course and career, supported and advised by the Careers and Employability Services team and the Professional Placement Module Leader. Workshops are provided on CV creation, interview techniques and placement searching, with drop-in sessions to provide additional support. In order to take the Professional Placement module, students need to have arranged a placement, approved by the Course Leader, by the end the preceding teaching block. Students undertake the placement

either before the final module, or following the completion of the final module. During the placement students will be supervised (online) by a tutor who, if possible, will visit during the placement. Depending on the location of the placement, meetings might take place face-to-face or via platforms such as Skype. The placement module will make use of the Virtual Learning Environment (VLE) Canvas for communication and dissemination of information between students and staff as well as making online learning materials available. Students will be required to keep a critical diary of the tasks and duties undertaken during their placement, to provide a framework for a reflective essay and work placement report, which is submitted for assessment at the end of the placement. They will also assemble and collate samples of work and evidence of achievement produced during their placement which will also be submitted for assessment. Placement providers will be asked to appraise the students' work and this feedback will be made available to the student. The placement module will be assessed on a pass/fail basis.

This integrated placement(s) provides students with a valuable opportunity to apply and develop their knowledge and skills in a professional working environment, enabling them to deepen their knowledge of the industry, develop their self-confidence, and strengthen their CV. Students undertaking placement activities are in a stronger position to gain the skills and experience which are valued by employers.

B. Aims of the Programme

The general aims of the Programme are:

- to offer students the opportunity to study modern European philosophy and contemporary art theory to an advanced level within a taught programme.
- to enhance students' knowledge of the main lines of inquiry in contemporary art theory, informed by an awareness of their grounding in the development of the post-Kantian philosophical tradition.
- to enable students to develop an advanced understanding of current debates in the field of contemporary art theory.
- to foster students' ability to develop general philosophical skills, both in class discussions and in individual written work, of interpretation, analysis, criticism and argument.
- to provide training in advanced research skills and the use of research resources, both physical and electronic.
- to develop students' ability to construct a complex argument and to express that argument in clear and accurate English prose.
- to provide students with a solid foundation for further postgraduate research or professional development.
- The 2-year programme with integrated placement(s) also provides students with an opportunity to enhance their professional skills, preparing them for higher levels of employment, further study and lifelong learning

This programme allows for three exit qualifications: Masters (180 credits), post-graduate Diploma (120 credits), and post-graduate Certificate (60 credits).

C. Intended Learning Outcomes

The programme provides opportunities for students to develop and demonstrate knowledge and understanding, skills and other attributes in the following areas. Where appropriate, the programme outcomes are referenced to the QAA subject benchmarks for Philosophy* and the Frameworks for Higher Education Qualifications of UK Degree-Awarding Bodies (2014), and relate to the typical student.

* There is no benchmark statement relating specifically to Philosophy at Level 7. Where appropriate, we are guided by the most recent QAA benchmark statement for Philosophy at Honours level.

PROGRAMME SPECIFICATION

MA Aesthetics and Art Theory – 2018-19

Programme Learning Outcomes					
	<u>Knowledge and Understanding</u>		<u>Intellectual skills – able to:</u>		<u>Subject Practical skills</u>
	<u>On completion of the course students will have advanced knowledge and understanding of:</u>		<u>On completion of the course students will be able to:</u>		<u>On completion of the course students will be able to:</u>
<u>A1</u>	<u>The main concepts introduced and discussed by Immanuel Kant in his <i>Critique of Judgement</i>, in the light of recent and current critical debates</u>	<u>B1</u>	<u>Interpret and analyse complex and difficult philosophical texts, informed by current critical practice</u>	<u>C1</u>	<u>Work independently and manage their time effectively</u>
<u>A2</u>	<u>The formative role of Kant's <i>Critique of Judgement</i> from the nineteenth to twenty-first-century aesthetic tradition</u>	<u>B2</u>	<u>Consider and critically examine unfamiliar ideas and terminology</u>	<u>C2</u>	<u>Prepare and deliver effective oral presentations of their work</u>
<u>A3</u>	<u>The main ideas and arguments of two or more major thinkers within the post-Kantian tradition of European philosophy, as presented in canonical texts</u>	<u>B3</u>	<u>Recognize methodological errors, rhetorical devices and unnoticed assumptions</u>	<u>C3</u>	<u>Locate appropriate electronic and physical research resources and plan a programme of library-based research appropriate to an MA dissertation</u>
<u>A4</u>	<u>The distinctive concepts, features and modes of argument and presentation of contemporary art theory.</u>	<u>B4</u>	<u>Construct and sustain complex arguments about matters of a fundamental and abstract nature, and express them in clear and accurate English prose.</u>	<u>C4</u>	<u>Organise and sustain wide-ranging research over a period of time and to structure and present a complex argument in a coherent fashion</u>

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In addition to the programme learning outcomes identified overleaf, the programme of study defined in this programme specification will allow students to develop a range of Key Skills as follows:

<u>Key Skills</u>						
<u>Self-Awareness Skills</u>	<u>Communication Skills</u>	<u>Interpersonal Skills</u>	<u>Research and information Literacy Skills</u>	<u>Numeracy Skills</u>	<u>Management & Leadership Skills</u>	<u>Creativity and Problem Solving Skills</u>
<u>Take responsibility for own learning and plan for and record own personal development</u>	<u>Express ideas clearly and unambiguously in writing and the spoken work</u>	<u>Work well with others in a group or team</u>	<u>Search for and select relevant sources of information</u>	<u>Collect data from primary and secondary sources and use appropriate methods to manipulate and analyse this data</u>	<u>Determine the scope of a task (or project)</u>	<u>Apply scientific and other knowledge to analyse and evaluate information and data and to find solutions to problems</u>
<u>Recognise own academic strengths and weaknesses, reflect on performance and progress and respond to feedback</u>	<u>Present, challenge and defend ideas and results effectively orally and in writing</u>	<u>Work flexibly and respond to change</u>	<u>Critically evaluate information and use it appropriately</u>	<u>Present and record data in appropriate formats</u>	<u>Identify resources needed to undertake the task (or project) and to schedule and manage the resources</u>	<u>Work with complex ideas and justify judgements made through effective use of evidence</u>
<u>Organise self effectively, agreeing and setting realistic targets, accessing support where appropriate and managing time to achieve targets</u>	<u>Actively listen and respond appropriately to ideas of others</u>	<u>Discuss and debate with others and make concession to reach agreement</u>	<u>Apply the ethical and legal requirements in both the access and use of information</u>	<u>Interpret and evaluate data to inform and justify arguments</u>	<u>Evidence ability to successfully complete and evaluate a task (or project), revising the plan where necessary</u>	
<u>Work effectively with limited supervision in unfamiliar contexts</u>		<u>Give, accept and respond to constructive feedback</u>	<u>Accurately cite and reference information sources</u>	<u>Be aware of issues of selection, accuracy and uncertainty in the collection and analysis of data</u>	<u>Motivate and direct others to enable an effective contribution from all participants</u>	
		<u>Show sensitivity and respect for diverse values and beliefs</u>	<u>Use software and IT technology as appropriate</u>			

D. Entry Requirements

The minimum entry qualifications for the programme are:

From BA: Normally a good (2:1 or upper-second class honours, or better) undergraduate degree or its equivalent in a relevant or related subject (including but not limited to philosophy, history, politics, cultural studies, fine art and visual culture, modern languages). Applicants with other kinds of qualifications will be considered on an individual basis.

A minimum, overall IELTS score of 6.5 or equivalent is required for those for whom English is not their first language.

E. Programme Structure

This programme is offered in full-time, part-time and ‘with professional placement’ mode, and leads to the award of MA Aesthetics and Art Theory (interim awards: PG Cert Aesthetics and Art Theory; PG Dip Aesthetics and Art Theory).

Entry is normally at level 7 with BA or equivalent qualifications (See section D). Transfer from a similar programme is possible at level 7 with good passes in comparable level 7 modules – but is at the discretion of the course team. Intake is normally in September.

E1. Professional and Statutory Regulatory Bodies

N/A

E2. Work-based learning

Work placement is an integral part of the 2-year programme and students will receive support from the award winning Careers and Employability Services team.

While it is the responsibility of individual students to secure appropriate placements, the Careers and Employability Services team offers each student support at all stages of the application process, including writing CVs, completing application forms, participating in mock interviews, assessment centre activities and psychometric tests. Sourcing and applying for placement(s) gives students the opportunity to experience a competitive job application process.

The experience of the work placement period enables students to apply their learning in the professional work environment, to reflect upon their own personal experience of working in an applied setting, to focus on aspects of this experience that they can clearly relate to their prior learning, and to evaluate the relationships between academic skills and employers’ expectations. Students will be assessed during and at the end of this period, through a portfolio of work, which will be marked as pass/fail.

E3. Outline Programme Structure

The degree consists of 180 credits, four modules of 30 credits each and a final dissertation of 60 credits. All students will be provided with the University regulations. Full details of each module will be provided in module descriptors and student module guides.

Students on the 2-year programme (with integrated placement) must complete all modules except the final ‘capstone project’ module, by the end of TB2, and then work in their placement(s) for a

maximum of 12 months. The student should confirm that their placement opportunity is available by the end of May, and the course team will confirm whether this is acceptable within two weeks. Students on placement(s) must complete a portfolio assessment which includes a reflection on how they have applied the skills they have developed during the previous year, within a professional working environment.

Level 7					
Compulsory modules	Module code	Credit Value	Level	Teaching Block	Pre-requisites
<u>Kant and the Aesthetic Tradition</u>	<u>PH7701</u>	<u>30</u>	<u>7</u>	<u>1</u>	
<u>Dissertation</u>	<u>PH7001</u>	<u>60</u>	<u>7</u>	<u>2 and 3</u>	<u>PH7701</u>
Option modules					Pre-requisites
<u>Romantic Philosophy of Art</u>	<u>PH7702</u>	<u>30</u>	<u>7</u>	<u>1</u>	<u>None</u>
<u>Art Theory: Modernist, Avant-garde, Contemporary</u>	<u>PH7703</u>	<u>30</u>	<u>7</u>	<u>2</u>	<u>None</u>
<u>Plasticity and Form</u>	<u>PH7704</u>	<u>30</u>	<u>7</u>	<u>2</u>	<u>None</u>
<u>Philosophy of Art History</u>	<u>PH7705</u>	<u>30</u>	<u>7</u>	<u>1</u>	<u>None</u>
<u>Kant and His Legacy</u>	<u>PH7801</u>	<u>30</u>	<u>7</u>	<u>1</u>	<u>None</u>
<u>Hegel and His Legacy</u>	<u>PH7802</u>	<u>30</u>	<u>7</u>	<u>1</u>	<u>None</u>
<u>Nietzsche and Heidegger</u>	<u>PH7803</u>	<u>30</u>	<u>7</u>	<u>2</u>	<u>None</u>
<u>Recent French Philosophy</u>	<u>PH7804</u>	<u>30</u>	<u>7</u>	<u>2</u>	<u>None</u>
<u>Topics in Modern European Philosophy</u>	<u>PH7805</u>	<u>30</u>	<u>7</u>	<u>2</u>	<u>None</u>
<u>Political Philosophy</u>	<u>PH7806</u>	<u>30</u>	<u>7</u>	<u>2</u>	<u>None</u>
<u>Critique, Practice, Power</u>	<u>PH7901</u>	<u>30</u>	<u>7</u>	<u>1</u>	<u>None</u>
<u>Recent Italian Philosophy</u>	<u>PH7903</u>	<u>30</u>	<u>7</u>	<u>1</u>	<u>None</u>
<u>German Critical Theory</u>	<u>PH7904</u>	<u>30</u>	<u>7</u>	<u>2</u>	<u>None</u>
<u>Marx and His Legacy</u>	<u>PH7905</u>	<u>30</u>	<u>7</u>	<u>2</u>	<u>None</u>
<u>Philosophy and Psychoanalysis</u>	<u>PH7906</u>	<u>30</u>	<u>7</u>	<u>2</u>	<u>None</u>
<u>Professional Placement</u>	<u>HU7100</u>	<u>120</u>	<u>7</u>	<u>TB3 (Yr 1) and TB1&TB2 (Yr 2)</u>	

Students must choose ONE of the following modules: PH7702 Romantic Philosophy of Art, PH7703 Art Theory: Modernist, Avant-garde, Contemporary, PH7704 Plasticity and Form, PH7705 Philosophy of Art History.

Students exiting the programme with 60 credits are eligible for the award of PgCert in Aesthetics and Art Theory.

Students exiting the programme with 120 credits are eligible for the award of PgDip in Aesthetics and Art Theory.

NB Not all modules run every year.

F. Principles of Teaching, Learning and Assessment

The Programme has been designed to take account of the KU Curriculum Design principles. The Programme uses a range of teaching and learning methods that encourage students' active engagement throughout. Teaching and learning methods are designed to suit the content and learning outcomes of the modules. Lectures are knowledge focussed, explaining core aspects of the

syllabus and its intellectual context, as well as offering models of interpretation and commentary. Seminar discussions explore the understanding of set texts and lecture materials and develop skills of exegesis, argumentation and oral presentation. Brief (formatively assessed) oral presentations help students develop their own understanding of the essentials of particular texts and topics and help students gain confidence in public speaking and response. As both presenters and respondents students engage in peer review and develop critical questioning skills. Individual and group tutorials offer opportunities to discuss essay topics and any issues students may find particularly difficult on the modules, and are used to receive detailed feedback on assessed written work. Individual and group tutorials are arranged with module tutors; on core modules and some option modules doctoral Teaching Assistants (CRMEP PhD students) also run tutorials. Students also learn autonomously through independent study of set texts, library research, peer engagement and discussion both inside and outside the class, and through individual writing. Development of academic skills is threaded throughout the whole course.

The regular extra curricular research events schedule is also partly designed to complement and extend formal teaching and learning on the MA. (See <http://fass.kingston.ac.uk/research/crmepevents/>). Research seminars, lectures, conferences and workshops include topics relevant to the MA and offer students the opportunity to engage with a wider national and international research community. An annual CRMEP Graduate Conference, organised by a team of PhD and MA students, allows students to participate in organisational and decision-making processes. Students who present papers at the conference also benefit from the experience of speaking and responding to a large public audience, and from further formative peer and faculty assessment. As well as formal class contact in lectures, seminars and tutorials, the extra-curricular events programme also allows for and encourages significant, informal staff-student contact as part of the enhanced learning and teaching environment. On-line learning technologies are also used throughout the course, to communicate with students, to make learning materials available to students and to foster on-line learning where appropriate.

Assessment in all modules (including the Dissertation module) is both formative and summative. All 30-credit modules are formatively assessed (by staff and peers) through seminar discussions and students' oral presentations. 30-credit modules are also summatively assessed either through a 1500-word written exercise and a 3500-4000-word essay or with one 5000-6000-word essay alone. The 1500-word written exercise is designed to ensure that students have grasped the basic philosophical content in the early stages of each module; to identify, early on, any problems that individual students may need to address and to identify the means to do so; to identify and devise strategies for building on individual student's strengths; and to build confidence and skills in academic writing. Individual feedback tutorials in which the written exercise are discussed also allow staff and teaching assistants to provide formative guidance and advice in preparation for the essay. The longer essays gives students the opportunity to explore topics in greater depth. Students take advantage of pre-essay tutorials to discuss and receive formative advice on essay plans; the summative assessment of essays includes written comments that provide formative guidance on skills for the preparation of the dissertation. In summative assessments the emphasis falls squarely on the development of the ability to demonstrate comprehension and to communicate aspects of difficult topics in writing. This is a major transferable skill and intense preparation for any further postgraduate (for example doctoral) study.

Formative assessment and peer engagement are also prominent features of the four Research Skills seminars which are included as part of the Dissertation module PHM7001. Seminars cover: i) accessing and using library and electronic resources, with the Philosophy Subject Librarian; ii) writing, composition and editing skills; iii) preparation of the dissertation proposal and writing the dissertation; and iv) oral presentations by students on agreed dissertation topics. The first three of these seminars comprise formal skills teaching; in the fourth seminar students practice and extend

oral presentation and response skills, receiving and participating in formative peer assessment, as well as receiving feedback from staff. In the Dissertation module, students apply the skills and knowledge acquired during the taught phase of the degree to the preparation of an extended piece of research-based writing (12–15000 words). Students are assigned an appropriate supervisor from the Philosophy team to guide them through this largely self-directed phase of the degree. Students meet regularly with their supervisor while preparing their dissertation to discuss the scope and intellectual content of the dissertation, research strategy and, as appropriate, early drafts of parts of their dissertation.

G. Support for Students and their Learning

In addition to structured teaching, guidance is available for students throughout the year through the provision of specific pre-set Office Hours, during which all members of the teaching staff are available for consultation with students. Students are encouraged to meet with teaching staff at such times, and at other times by appointment, for individual tutorial sessions. Students are also assigned a personal tutor, normally the tutor teaching the core module or the Philosophy Director of Studies. In addition to help and advice from module tutors the personal tutor tracks tutees' achievement, makes any appropriate recommendation for further support, and acts as the first contact for academic support and advice of all kinds. Provision is made for students to change personal tutor if necessary.

Students are supported by:

- A Module Leader for each module
- A Course Leader to help students understand the programme structure
- Personal Tutors to provide academic and personal support
- A placement tutor to give general advice on placements
- Technical support to advise students on IT and the use of software
- A designated Course Administrator
- An induction week at the beginning of each new academic session
- Staff Student Consultative Committee
- VLE/Canvas – a versatile online interactive intranet and learning environment accessible both on and off-site
- Lynda.com – an online platform offering self-paced software tutorials
- A substantial Study Skills Centre that provides academic skills support for both UG and PG students
- Student support facilities that provide advice on issues such as finance, regulations, legal matters, accommodation, international student support etc.
- A Student Achievement Officer who provides pastoral support
- Support for students with disabilities
- The Union of Kingston Students
- Careers and Employability Services Team, who will provide support for students prior to undertaking work placement(s).

H. Ensuring and Enhancing the Quality of the Course

The University has several methods for evaluating and improving the quality and standards of its provision. These include:

- External Examiners
- Boards of Study with student representation
- Annual Monitoring and Enhancement
- Periodic review undertaken at subject level
- Student evaluation including MEQs
- Moderation policies
- Feedback from employers

I. Employability Statement

The MA in Aesthetics and Art Theory degree fosters a range of skills highly desirable to employers, such as high level communication skills, the capacity to deal effectively with substantial quantities of complex information, skills in analysis, critical and creative thinking, research skills, self-management and the capacity to work to deadlines, meticulousness in written presentation, the ability to work both independently and constructively with others.

While some MA in Aesthetics and Art Theory students will be studying to begin or enhance a career path in teaching, research or in the creative industries, other graduates will go into a variety of careers, including public policy, media/journalism, publishing, arts administration, management, marketing, leisure and tourism, IT and a variety of public service and therapeutic fields. For those interested in further research, the course provides an excellent foundation for MPhil/PhD level study in related fields.

The 2-year integrated work placement programme is designed to provide students with enhanced opportunities for securing professional employment at the end of their degree, providing skills and experience that employers are looking for in their work force. These are supported by the services of the Careers and Employability Services team, providing drop-in and scheduled events to support students in the preparation of CVs, applications, and preparation for interviews and assessment centres.

Achievements of recent alumni of the Programme can be seen at <http://fass.kingston.ac.uk/research/crmep/alumni/>

J. Approved Variants from the Undergraduate Regulations

N/A

K. Other sources of information that you may wish to consult

Course page on the KU website

<https://www.kingston.ac.uk/postgraduate-course/aesthetics-art-theory-ma/>

Development of Programme Learning Outcomes in Modules

This map identifies where the field/course learning outcomes are **summatively** assessed across the modules for this field/course. It provides an aid to academic staff in understanding how individual modules contribute to the field/course aims, a means to help students monitor their own learning, personal and professional development as the field/course progresses and a checklist for quality assurance purposes.

		Level 7																
Module Code		PH7001	PH7701	PH7702	PH7703	PH7704	PH7705	PH7801	PH7802	PH7803	PH7804	PH7805	PH7806	PH7901	PH7906	PH7903	PH7904	PH7905
Programme Learning Outcomes	Knowledge & Understanding	A1	S	S														
		A2	S	S	S	S	S											
		A3	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
		A4	S	S	S	S	S											
	Intellectual Skills	B1	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
		B2	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
		B3	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
		B4	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
	Practical Skills	C1	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
		C2	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
		C3	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
		C4	S															

Students will be provided with formative assessment opportunities throughout the course to practise and develop their proficiency in the range of assessment methods utilised.

Technical Annex

<u>Final Award(s):</u>	<u>MA Aesthetics and Art Theory</u>
<u>Intermediate Award(s):</u>	<u>Postgraduate Diploma in Aesthetics and Art Theory, Postgraduate Certificate in Aesthetics and Art Theory</u>
<u>Minimum period of registration:</u>	<u>1 year full-time, 2 years full-time (with Professional Placement); 2 years part-time</u>
<u>Maximum period of registration:</u>	<u>2 years full-time, 3 years full-time (with Professional Placement); 4 years part-time</u>
<u>FHEQ Level for the Final Award:</u>	<u>Masters</u>
<u>QAA Subject Benchmark:</u>	<u>There is currently no benchmarking statement specific to Philosophy at postgraduate level.</u>
<u>Modes of Delivery:</u>	<u>Full-time, Part-time and 'with Professional Placement'</u>
<u>Language of Delivery:</u>	<u>English</u>
<u>Faculty:</u>	<u>KSA</u>
<u>School:</u>	<u>Arts, Culture and Communication</u>
<u>Department:</u>	<u>Humanities</u>
<u>UCAS Code:</u>	<u>N/A</u>
<u>Course/Route Code:</u>	

BIOLOGICAL SCIENCES: Neuroscience

Definition of the Hypothalamic GnRH Pulse Generator in mice

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~~Short title: Definition of GnRH pulse generator~~

~~Key words: episode generator, fertility, GCaMP, GnRH, hypothalamus, kisspeptin, optogenetics, photometry, pulsatility.~~

Abstract

The pulsatile release of luteinizing hormone (LH) is critical for mammalian fertility. However, despite several decades of investigation, the identity of the neuronal network generating pulsatile reproductive hormone secretion remains unproven. We use here a variety of optogenetic approaches in freely behaving mice to evaluate the role of the arcuate nucleus kisspeptin (ARN^{KISS}) neurons in LH pulse generation. Using GCaMP6 fibre photometry, we find that the ARN^{KISS} neuron population exhibits brief (~1 min) synchronized episodes of calcium activity occurring as frequent as every 9 min in gonadectomized mice. These ARN^{KISS} population events were found to be near-perfectly correlated with pulsatile LH secretion. The selective optogenetic activation of ARN^{KISS} neurons for 1 min generated pulses of LH in freely behaving mice, whereas inhibition with archaerhodopsin for 30 min suppressed LH pulsatility. Experiments aimed at re-setting the activity of the ARN^{KISS} neuron population with halorhodopsin were found to re-set on-going LH pulsatility. These observations indicate the ARN^{KISS} neurons as the long-elusive hypothalamic pulse generator driving fertility.

Significance statement

Neural networks located in the hypothalamus are responsible for generating ultradian patterns of hormone secretion that control a wide variety of functions. How these neural networks generate pulsatile hormone secretion remain unknown. We report here that a population of hypothalamic kisspeptin neurons represent the gonadotropin-releasing hormone (GnRH) pulse generator. These cells have the remarkable ability to generate synchronized GnRH secretion every 9 minutes to drive pulsatile gonadotropin hormone secretion in the blood. These observations indicate the arcuate kisspeptin neurons as the origin of reproductive hormone pulsatility in mice and offer the prospect of better understanding and manipulating fertility in the clinic.

body

Endocrine signaling from the pituitary gland occurs primarily as pulsatile hormone secretion. For most hormones, this pulsatility is thought to be generated by episode generators located within the hypothalamus and pituitary gland (1, 2). However, with few exceptions, it has been extremely difficult to determine the identity and nature of these neuroendocrine pulse generators (3, 4). Although the critical importance of pulsatile hormone secretion for reproduction was discovered in the 1970s (5, 6), the identity of the so-called “gonadotropin-releasing hormone (GnRH) pulse generator” (7, 8) responsible for pulsatile gonadotropin secretion remains unknown.

The discovery of the critical importance of kisspeptin signalling within the brain of humans and rodents for fertility (9–11) has suggested a potential role for kisspeptin signaling in the generation of pulsatile hormone secretion. Humans and rodents with global deletions or mutations in kisspeptin or the kisspeptin receptor fail to exhibit normal pulsatile luteinizing hormone (LH) secretion (9, 10, 12, 13). Of the two kisspeptin neuron populations that innervate GnRH neurons, particular attention has focussed upon the arcuate nucleus kisspeptin (ARN^{KISS}) neurons in terms of pulse generation (14, 15). Investigators have shown that ARN^{KISS} neurons co-express a variety of different neurotransmitters including glutamate, neurokinin B and dynorphin (15, 16), for which these cells also express functional receptors (17, 18). Alongside anatomical evidence, it has been suggested that the ARN^{KISS} neurons may exhibit synchronized behaviour that could intermittently activate GnRH neurons and, thus, act as a pulse generator (15, 16, 19). This attractive hypothesis remains unproven.

We employ here a range of optogenetic approaches in freely behaving mice to evaluate whether the ARN^{KISS} neurons represent the GnRH pulse generator. Using GCaMP fiber photometry, we observed brief repetitive episodes of elevated calcium within the ARN^{KISS} neuron population and find a near-perfect correlation of these events with pulsatile LH secretion. Mimicking these brief synchronization events through selective channelrhodopsin (ChR2) activation of ARN^{KISS} neurons was sufficient to generate pulsatile LH secretion while optogenetic inhibition of ARN^{KISS} neurons suppressed pulsatile LH secretion. Further, re-setting the activity of the ARN^{KISS} neuron population was found to re-set pulsatile LH secretion. Together these observations indicate that the ARN^{KISS} neurons are the GnRH pulse generator.

RESULTS

ARN^{KISS} neurons exhibit episodic activity in vivo

We used fiber photometry (20) to examine whether ARN^{KISS} neurons exhibited intermittent synchronous activity in awake freely behaving mice. We targeted the calcium reporter GCaMP6s to ARN^{KISS} neurons by bilateral injection of an AAV9-CAG-Flex-GCaMP6s into the ARN of male KISS1-Cre;tdTomato reporter mice (21). Immunohistochemical analyses showed that GCaMP6 was expressed in kisspeptin neurons throughout the ARN with 51–67% of ARN^{KISS} neurons expressing GCaMP6 depending on rostro-caudal level (Suppl Fig.1). Dual labeled cells represented 89±6% to 96±6% of all GCaMP6 expressing cells in the ARN (Suppl Fig.1). Acute brain slice experiments using

cell-attached recording combined with GCaMP6 calcium imaging showed that GCaMP6 faithfully reports on the electrical activity of ARN^{KISS} neurons in these mice (6 cells from 4 mice; Fig.1B).

We next injected a group of KISS1-Cre male mice with GCaMP6 AAV and implanted a 400µm diameter optical fiber just above the ARN (Fig.1A). Mice were gonadectomised (GDX) to generate a state in which the GnRH pulse generator is operating at its highest frequency. After three weeks of daily handling and habituation, the mice were connected to the patch cord of a fibre photometry system and calcium signals measured for up to 4h in freely behaving mice. Abrupt increases in calcium signal were detected at approximately regular intervals of 8.9 ± 0.5 min (range; 4.5–20.7 min) in all mice examined (n=7; Fig.1C). Histological analysis showed that all optical fibers were located above the middle to caudal aspect of the ARN. Photometry sampling revealed a consistent temporal dynamic for calcium events with an initial rapid increase over 19.6 ± 2.4 s followed by a slower decrease that, together, lasted for 74.2 ± 6.6 s (16 events from 7 mice; Fig.1D,E).

To demonstrate that the GCaMP6 signal reflects the activity of ARN^{KISS} neurons, we co-injected Cre-dependent AAVs for both GCaMP6s and hM3Dq (22) into the ARN of GDX KISS1-Cre mice. Three to 4 weeks later, calcium imaging experiments in acute brain slices demonstrated that addition of 1–10µM clozapine N-oxide (CNO) to the bathing medium resulted in a marked and prolonged increase in both firing and intracellular calcium concentrations ($[Ca^{2+}]_i$) in all ARN^{KISS} neurons (4 cells dual recorded, 38 cells in total, 3 mice; Fig.1F). KISS1-Cre mice injected with GCaMP6s and hM3Dq were then prepared with a photometry fibre directed at the ARN. Peripheral administration of CNO (i.p., 2mg/kg) to freely behaving mice was found to rapidly (within 5 min) enhance basal $[Ca^{2+}]_i$ by $33 \pm 8\%$ and increase the frequency of calcium events by $100 \pm 52\%$ ($p=0.04$, paired t test; n=4; Fig.1G).

These studies indicate that ARN^{KISS} neurons exhibit episodes of synchronized activity every ~9 min. This was surprising as previous estimates of GnRH pulse generator activity in GDX mice, obtained from measuring plasma luteinizing hormone (LH) levels (23), indicated a GnRH/LH pulse interval of >20 min. We next examined the relationship of ARN^{KISS} neuron calcium events to pulsatile LH secretion by taking 3 min tail tip blood samples for a 60–80 min period while also monitoring calcium signals. This revealed that calcium events were almost perfectly correlated with increments in LH secretion; 94% of calcium events were followed by a >10% increase in LH concentration. No increases in LH were detected without a preceding calcium event (60 LH pulses in 7 mice; Fig.1H & I). The interval from the peak of the calcium event to the peak of the LH pulse was highly consistent across all animals being 4.00 ± 0.14 min (n=7). A significant temporal relationship existed between calcium and LH peaks ($P<0.05$, repeated measures ANOVA), while no such relationship existed between randomly selected LH samples and calcium peaks (Suppl.Fig.2).

Brief activation of ARN^{KISS} neurons generates LH pulses in vivo

We have previously demonstrated that optogenetic activation of ARN^{KISS} neurons for >2 min at 10Hz generates pulse-like increments in LH secretion in anaesthetized mice (24). However, the fibre photometry studies above clearly demonstrate that ARN^{KISS} neuron calcium events last for only ~1 min. Hence, we examined whether brief 1 min periods of ARN^{KISS} neuron activation were sufficient to evoke an LH pulse in awake freely behaving mice:

Channelrhodopsin (ChR2) was targeted to ARN^{KISS} neurons by unilateral injection of a AAV9-EF1-dflox-hChR2-mCherry into the ARN of intact male and female KISS1-Cre mice (24). Intact mice exhibit an LH pulse every 90–180 min (12, 23) thereby providing a long window of basal LH secretion for optogenetic stimulation experiments. Activation of ARN^{KISS} neurons for 1 min (10Hz, 473nm light, 5mW) was found to reliably evoke pulse-like increments in LH secretion in conscious male (n=5; Fig.2A–C) and female (n=6; Fig.2D,E) mice and this was repeatable at 30 min intervals. Compatible with sex differences in endogenous LH pulse amplitude (12, 23), optogenetic activation of ARN^{KISS} neurons evoked 0.5–1ng amplitude LH pulses in females (Fig.2D,E) and ~4ng/ml amplitude pulses in males (Fig.2B,C). The profile of LH secretion evoked by 10Hz 1min stimulation was very similar to the profile of endogenous LH pulses observed in intact male mice (Fig.2F).

The inhibition of ARN^{KISS} neuron firing suppresses pulsatile LH secretion in vivo

To examine the necessity of ARN^{KISS} neurons for pulsatile LH secretion we employed an optogenetic archaerhodopsin (ArchT) mediated silencing approach (25, 26). Bilateral injections of a Cre-dependent ArchT-tdTomato AAV were made into the ARN of adult female KISS1-Cre;GFP reporter mice that were also GDX at the time of surgery to generate high frequency pulsatile LH secretion. This resulted in 60–69% of ARN^{KISS} expressing ArchT (Fig.3B & Suppl. Fig.3), representing 77–83% of all ArchT-expressing neurons depending on rostro-caudal location within the ARN (Suppl Fig.3). The use of a GFP reporter for labelling kisspeptin neurons is not as efficient as the tdTomato reporter (GCaMP6 and eNPHR3.0 experiments) and, consequently, under-reports the number of ARN^{KISS} neurons (Suppl. Figs 1,3,4).

Cell-attached recordings from ARN^{KISS} neurons in acute brain slices prepared from ArchT AAV-injected KISS1-Cre mice demonstrated that 532nm green light promptly suppressed firing rates by 76±8% over the 5-min period with a gradual return to baseline firing when illumination was stopped (11 cells from 5 mice; Fig.3C,D). As ARN^{KISS} neurons exhibit very low levels of spontaneous electrical activity in the brain slice, 10–20nM neurokinin-B was added to the bathing medium to increase their firing (27).

In awake mice (n=6), bilateral green light activation of ArchT in ARN^{KISS} neurons for a 30-min period resulted in a 49% suppression of LH pulse frequency ($p<0.01$; repeated measures ANOVA with *post-hoc* Dunnett's tests), 65% decrease in the amplitude of pulses occurring during illumination ($p<0.05$; ANOVA with *post-hoc* Dunnett's tests) and a 49% decline in basal LH secretion (Fig.3E,F,I–K). Controls (n=9) included blue light illumination in four mice previously showing LH pulse inhibition with green light (Fig.3G), green light illumination of wild-type mice injected with the Cre-dependent ArchT AAV (Fig.3H) and Kiss-Cre mice without ArchT. In all cases, laser illumination had no effect on LH secretion (Fig.3L–N).

Re-setting the activity of ARN^{KISS} neurons re-sets pulsatile LH secretion in vivo

The above studies show that ARN^{KISS} neuron activity is both necessary and sufficient for pulsatile LH secretion in the mouse. However, it remains possible that these cells represent a relay of the pulse generator rather than being the episode generator themselves. In the course of developing the optogenetic methodology for this study we also examined halorhodopsin (eNpHR3.0) as a potential

inhibitory tool. Electrophysiological studies on eNpHR3.0-expressing ARC^{KISS} neurons revealed that 532nm green light evoked a mean $81 \pm 4\%$ inhibition, and often complete silencing, of NKB activated ARC^{KISS} neurons but then resulted in a marked rebound activation for up to 2 min when illumination was stopped (Fig.4C,D). This represented an ideal way to temporally re-set the activity of the ARC^{KISS} neuron population *in vivo*. As such, we undertook a second series of optogenetic inhibition studies using eNpHR3.0 to both confirm the ArchT result and also determine whether we could re-set LH pulsatility through manipulation of only the ARN^{KISS} neurons.

Bilateral injections of Cre-dependent eNpHR3.0 AAV were made into the ARN of adult GDX female KISS1-Cre mice. This resulted in 63-70% of ARC^{KISS} neurons expressing eNpHR3.0 (Fig.4B, Suppl Fig.4) with dual-labelled neurons accounting for up to 94% of all eNpHR3.0 cells depending on rostro-caudal location within the ARN (Suppl. Fig.3).

Bilateral green light activation of eNpHR3.0 in ARC^{KISS} neurons resulted in the suppression of pulsatile LH secretion for the 30-min duration of illumination (Fig.4E,F; n=8). This resulted in a 66% decrease in LH pulse frequency ($p < 0.05$; ANOVA with *post-hoc* Dunnett's tests) with 5 of 8 mice exhibiting a complete suppression of pulsatile LH release during the 30-min illumination period (Fig.4E,F). The remaining 3 mice exhibited a single pulse during this time with, overall, a 65% reduction in LH pulse amplitude ($p < 0.01$; ANOVA with *post-hoc* Dunnett's tests; Fig.4I). Control mice (n=7) including wild-type mice injected with eNpHR3.0 AAV and KISS1-Cre mice with off-target fiber optic placements, did not exhibit any changes in pulsatile or basal LH secretion during green light illumination (Fig.4G,H,L-N). Four mice with unilateral activation of eNpHR3.0 in ARN^{KISS} neurons did not exhibit any changes in LH (Suppl. Fig.5E).

As predicted, a robust LH pulse was generated immediately after termination of the green light illumination in eNpHR3.0 KISS1-Cre mice (Fig.4E,F,I), but not in controls (Fig.4G,H,L). KISS1-Cre mice also exhibited a re-setting of on-going LH pulsatility (Fig.4E,F). As observed with mean LH values (Fig.4I), the random timing of LH pulses in individual mice results in an averaged flat line before a fall in mean LH levels upon illumination and the initiation of an LH pulse when the laser is turned off. Subsequent to this evoked pulse, another was found to occur 35 min later in all mice (Fig.4I arrow). Control mice, and indeed ArchT KISS1-Cre mice (Fig.3I), do not show any relationship between LH pulsatility and green light termination (Fig.4G,H,L). For unknown reasons, the expression of eNpHR3.0 in ARC^{KISS} neurons generates mice with large, relatively slow LH pulses independent of any illumination. As such, LH pulses occur every 34 min in these mice (Fig.4K) and this is entirely consistent with the on-going 35 min pulse interval found after re-setting LH pulse frequency with eNpHR3.0 (Fig.4I).

In undertaking the eNpHR3.0 experiments, we generated cohorts of KISS1-Cre mice in which eNpHR3.0 was expressed throughout the ARN but optical fibers located bilaterally in either the rostral (n=5) or middle/caudal (N=7) ARN (Suppl. Fig.5). Whereas light activation of the middle/caudal ARN^{KISS} neurons was always effective in suppressing pulsatile LH secretion, illumination of the rostral ARN^{KISS} had no (Suppl. Fig.5F), or only partial (Suppl. Fig.5G), effects on pulsatile LH secretion.

DISCUSSION

We demonstrate here that the ARN^{KISS} neurons are almost certainly the long-elusive hypothalamic GnRH pulse generator. These neurons exhibit episodes of synchronized activity lasting ~1min that are almost perfectly correlated with pulsatile LH secretion. Consistent with this, the activation of ARN^{KISS} neurons for 1 min reliably generated pulses of LH secretion. Using two inhibitory optogenetic strategies, we show that the selective inhibition of ARN^{KISS} neurons activity results in the suppression of pulsatile LH secretion. Furthermore, the re-setting of ARN^{KISS} neuron activity is found to re-set pulsatile LH secretion.

The calcium events detected with fiber photometry very likely represent the synchronized electrical activity of the ARN^{KISS} neurons. In brain slice experiments, we found that GCaMP6 reliably reported ARN^{KISS} neuron firing. Furthermore, co-expression of GCaMP6 and hM3Dq in ARN^{KISS} neurons demonstrated that the CNO-induced activation of firing in ARN^{KISS} neurons was associated with large increases in $[Ca^{2+}]_i$ within individual ARN^{KISS} neurons. We noted that, *in vivo*, CNO raised the basal calcium signal and doubled the frequency of synchronized calcium events. This shows that the ARN^{KISS} neuron population can maintain an episodic synchronized output in the presence of persistent Gq-protein-coupled receptor activation. This did not occur in the acute brain slice where CNO evoked a long-lasting activation of both firing and $[Ca^{2+}]_i$ in individual ARN^{KISS} neurons with no evidence of episodic activity. This possibly results from the absence of network activity in the acute brain slice where the deafferented ARN^{KISS} neurons show little spontaneous activity and no episodic or synchronous firing (27, 28).

The ARN^{KISS} neuron population exhibits robust increases in calcium approximately every 9 min. This was unexpected given that prior assessments of pulse generator activity had indicated LH pulse interval to be >20 min (29). Nevertheless, we find here that 94% of synchronized calcium events are followed by an increment in LH secretion in gonadectomized mice. Often these changes in LH did not occur as a classical pulse but could be observed, for example, as an increment in LH on the downward shoulder of a prior LH pulse. These events are not detected by current pulse analysis algorithms and indicate the need for the further development of LH pulse analysis methodologies. In terms of ARN function, it is intriguing to find a 10-min neuronal oscillator embedded amongst ARN neural circuitries subserving longer term function such as feeding initiation (30). With evidence for kisspeptin inputs into these feeding circuits and others (31), it will be intriguing to examine the impact of this fast oscillator on other ARN outputs in future studies.

The selective inhibition of ARN^{KISS} neuron activity with both ArchT and eNpHR3.0 optogenetic strategies suppressed pulsatile LH secretion. We found that the eNpHR3.0 approach generated a stronger inhibition of ARN^{KISS} neuron firing *in vitro* and this correlated with a more robust inhibition of LH pulsatility compared with ArchT. It is likely that a sub-population of the ~1,000 ARN^{KISS} neurons in the mouse (21) are required for pulse generation. Only ~70% of ARN^{KISS} neurons expressed ArchT or eNpHR3.0 in our experiments and fibre optic locations in the rostral ARN were ineffective in suppressing pulsatile LH secretion. This raises the possibility that a sub-population of kisspeptin neurons located in the mid-caudal ARN may comprise the GnRH pulse generator. How few ARN^{KISS} neurons may be required for pulsatility is not known but a 30% suppression of total *Kiss1* mRNA expression in the ARN does not appreciably alter LH pulsatility in rats (32). Unilateral suppression of ARN^{KISS} neuron activity was insufficient to alter LH pulsatility, whereas unilateral activation generated LH pulses. Contralateral projections between the bilateral ARN^{KISS} neuron networks may maintain their synchronization (18, 33), although a unilateral population is sufficient to generate an LH pulse.

Although the ChR2 and ArchT optogenetic studies demonstrated the necessity and sufficiency of ARN^{KISS} neurons for pulse generation, they did not establish whether this population was, itself, the pulse generator. The silencing followed by rebound activation of ARN^{KISS} after eNpHR3.0 activation provided a useful tool to explore this issue. Rebound activation has been reported following eNpHR, but not ArchT, activation in other neural populations and may result from chloride ion-based changes in GABA_A reversal potential (34). We predicted that if ARN^{KISS} neurons were simply a relay within the pulse generator, then eNpHR3.0-mediated synchronous activation would have no impact upon LH pulsatility beyond the initiated pulse. On the other hand, if the ARN^{KISS} neurons were the genesis of pulsatility this randomly timed activation should be able to re-set future pulsatile LH secretion. This was indeed found to be the case with synchronous eNpHR3.0 activation generating an LH pulse upon offset as well as a subsequent time-locked spontaneous LH pulse ~ 35 min later in every mouse. This also occurred regardless of when the stimulus was applied in relation to endogenous LH pulsatility. We were surprised to find that the expression of eNpHR3.0 in ARN^{KISS} neurons resulted in the slowing of LH pulse frequency to ~34 min independent of any light activation. Light-independent effects of eNpHR3.0 on neuronal behaviour have not been reported previously but, nevertheless, suggest the possibility that changes in ARN^{KISS} neuron chloride ion homeostasis may be one strategy for slowing pulse generator frequency.

Tracing studies in mice have demonstrated that ARN^{KISS} neurons only project to the distal projections of GnRH neurons in the vicinity of the median eminence (35). These distal GnRH neuron projections, termed dendrons, are unusual in that they display both dendrite and axon-like properties and are regulated directly by multiple neural inputs (36, 37). We show here that ARN^{KISS} neurons release kisspeptin for ~1min to generate an LH pulse and recent studies have shown that brief (10-60s) applications of kisspeptin to the median eminence increase intracellular calcium levels in GnRH neuron dendrons and evoke GnRH secretion independent of action potential firing in GnRH neurons (37, 38). This ARN^{KISS} → GnRH neuron dendron pathway is also likely to exist in primates as the closest associations between kisspeptin and GnRH fibers occur in the internal zone of the median eminence of the monkey (39) and kisspeptin levels within the ARN/median eminence fluctuate and can occur coincident with episodes of pulsatile GnRH secretion in this species (40). The ability of ARN^{KISS} neurons to target the merged distal projections of GnRH neurons around the median eminence abrogates the need for a pulse generator to project to the widely dispersed GnRH neuron cell bodies. This represents an atypical mechanism for rhythm generation in the brain (41) and emphasizes the diversity of mechanisms that the neuroendocrine hypothalamo-pituitary axes likely employ to drive the pulsatile hormone secretion.

In conclusion, we provide here robust evidence for the identity of the GnRH pulse generator proposed over three decades ago. It remains that the mechanisms responsible for the synchronization of ARN^{KISS} neurons are unknown. Similarly, it is unclear how gonadal steroids and neuronal inputs proposed to mediate multiple different homeostatic and external influences upon pulsatility (14) interact with the ARN^{KISS} neurons. It is expected that the further elucidation of both the intra- and inter-cellular mechanics of the ARN^{KISS} pulse generator will provide a robust framework for advancing strategies aimed at regulating LH pulse frequency in the clinic.

METHODS

Animals:

Adult littermate heterozygous *KISS1-Cre^{+/−}* or *KISS1-Cre^{−/−}* (21) mice, alone or crossed onto Cre-dependent reporter ROSA26-CAG-*τGFP* or Ai9-CAG-*tdTomato* lines (mixed 129S6Sv/Ev C57BL6 genetic background) were housed under a 12:12h lighting schedule (lights on at 6:00 A.M.) with *ad libitum* access to food and water. All experiments were approved by the University of Otago Animal Welfare and Ethics Committee.

Stereotaxic injections of AAVs:

Recombinant Cre-dependent AAVs encoding GCaMP6s (AAV2/9-CAG-FLEX-GCaMP6s-WPRE-SV40, 1.3×10^{13} mol/ml; U. Penn) ChR2 (AAV2/9-EF1-DIO-hChR2-(H134R)-mCherry-hGH, 5.8×10^{13} ; U. Penn), ArchT (AAV2/9-CAG-FLEX-ArchT-*tdTom*, 1.5×10^{13} ; U. North Carolina), eNpHR3.0 (AAV2/1-EF1-DIO-eNpHR3.0-EYFP, 2.8×10^{13} ; U. Penn) or hM3Dq (AAV2-hSyn-DIO-hM3D(Gq)-mCherry, 4×10^{12} ; U. North Carolina) were injected bilaterally into the mouse ARN. Animals were given Carprofen (5mg/kg b.w. s.c), anesthetized with 2% isoflurane, and placed in a stereotaxic apparatus. A custom-made bilateral Hamilton syringe apparatus holding two 1 μ l Hamilton syringes with 25 gauge needles held 0.9 mm apart was used to perform bilateral injections. Each syringe was filled with 1.0 μ l of AAV. For the GCaMP6 and ChR2 experiments, mice received a single bilateral injection of AAV 1.2mm posterior to bregma, and 6.0mm deep. The needles were lowered into place and after 3 min, 1 μ l of AAV injected over 10 min, and the needles left *in situ* for a further 10 min before being withdrawn. For the ArchT and eNpHR3.0 experiments mice received two bilateral injections (1 μ l per injection per hemisphere) at 1.0 and 1.2mm posterior to Bregma, and 5.7mm depth. Some mice received bilateral 1 μ l co-injections of GCaMP6s and hM3Dq AAVs.

Brain-slice electrophysiology and calcium imaging:

Electrophysiological recordings of ARN^{KISS} neurons were carried out as described previously (27). Briefly, coronal brain slices (250 μ m thick) were placed under an upright epifluorescence microscope and constantly perfused (1.5mL/min) with 95%O₂/5%CO₂ equilibrated 30°C artificial cerebrospinal fluid (aCSF; in mM): NaCl 120, KCl 3, NaHCO₃ 26, NaH₂PO₄ 1, CaCl₂ 2.5, MgCl₂ 1.2 and glucose 10. Neurons expressing GFP or *tdTomato* were first visualized by brief fluorescence illumination, and subsequently approached using infrared differential interference contrast optics. ARN^{KISS} neuron action potential firing was recorded in voltage clamp mode using the cell attached loose patch configuration. Recording electrodes (3–5 M Ω) were filled with aCSF including 10 mM HEPES and low resistance seals (10–30 M Ω) achieved. Electrophysiological signals were recorded using a Multiclamp 700B amplifier connected to a Digidata 1440A digitizer (Molecular Devices). Signals were low-pass filtered at 3 kHz before being digitized at a rate of 10 kHz. Signal acquisition was carried out with pClamp 10 (Molecular Devices). Spontaneous spikes were detected using the threshold-crossing method in pClamp 10. Action potential firing was monitored for >2 min before 5 min illumination with a 532nm laser (0.6mW, Ikecool) delivered through a fiber optic (200 μ m diameter, NA 0.37; Doric Lenses) placed above the surface of the slice, near the recorded cell. For chemogenetic experiments, action potential firing of ARN^{KISS} neurons was monitored for >3 minutes before CNO (1–10 μ M) was bath-applied for 10 minutes. Statistical comparisons were undertaken using the non-parametric Friedman tests with the sample size used in tests being the number of neurons.

Brain-slice calcium imaging was carried out as previously reported (42). In brief, brain slices (as above) were illuminated through a 40X immersion objective, using the xenon arc light source (300 W; filtered

by a GFP filter cube, excitation 460–480 nm, Olympus) and the shutter of a Lambda DG-4 (Sutter Instruments). Epifluorescence (485 nm long pass and emission 495–540 nm) was collected using a Hamamatsu ORCA-ER digital CCD camera. A focal plane including several fluorescent ARN^{KISS} somata was chosen and acquisitions (100 ms light exposure at 2 Hz for 10–20 min) started. For analysis, time-series of images were processed in ImageJ. Regions of interest comprised of individual in-focus fluorescent ARN^{KISS} somata with average fluorescence intensity measured in each frame. Fluorescence intensity data were analyzed using scripts written in R (<http://www.r-project.org/>) or using pClamp 10. Relative fluorescence changes were calculated using the following formula: $\Delta F/F = (F_t - F)/F * 100$ where F is the baseline fluorescence intensity calculated as the mean fluorescence intensity over a 2-min period preceding drug applications and F_t is the test fluorescence.

Fiber photometry:

Following AAV injection of GCaMP6s, an optical fiber (400µm diameter, 0.48 NA; Doric Lenses, Canada) was positioned immediately above the mid-caudal ARN and the mouse GDx. Mice were housed individually and allowed to recover for at least three weeks with daily handling and habituation to the fiber photometry recording set-up. Over a period of 3–10 weeks following surgery, fluorescence signals were recorded from freely behaving mice in their home cages. Mice injected with GCaMP6s and hM3Dq AAVs received a single injection of CNO (2mg/kg, s.c.) during the recording.

Fluorescence signals were acquired using a fiber photometry system based on a previous published design (20). All optical components were purchased from Doric Lenses. Excitation light was provided by violet (referred to as 405nm) and blue (referred to as 490nm) fiber-coupled LEDs which were sinusoidally modulated at 531 and 211Hz respectively. Each excitation line passed through an excitation filter (400–410nm and 460–490nm) and focused into a 400µm optic fiber which connected to the mouse. Emitted fluorescence was collected by the same fiber, passed through a 500–550nm emission filter and focused onto a photoreceiver (2151 Newport). Two GCaMP6s emission signals were recovered by demodulating the 405nm (531Hz) and 490nm (211Hz) signals. The 405nm signal reported the excitation/emission of GCaMP6s at its isosbestic point and was used as a background (non-calcium dependent) signal. The 490nm signal, reported calcium dependent GCaMP6s excitation/emission and was used to infer changes in $[Ca^{2+}]_i$. The signals were modulated/demodulated and recorded at 10Hz with custom software (Tussock Innovation, Otago, NZ). Signals from the 405nm control channel were subtracted from the 490 channel before analysis. Photometry data could either be collected continuously or in 5s on/15s off scheduled mode. The power output at the tip of the fiber was set at 50µW. Fluorescence signals (490–405) were collected and converted to $\Delta F/F$ (%) values as follows: $\Delta F/F = 100 * (F - F_b)/F$ where F_b was the basal fluorescence signal between events and F the recorded fluorescence.

To assess the relationship of calcium events with pulsatile LH secretion, freely behaving mice were attached to the fiber photometry system and LH secretion measured by obtaining 3µL blood samples every 3–6 min from the tail tip over a period of 60–80 min. Blood sampling and subsequent ELISA measurement of LH was undertaken as reported previously (12, 23). The assay sensitivity was 0.04ng/mL, with intra- and inter-assay coefficients of variation of 9.3% and 10.5%, respectively.

Optogenetic experiments:

Adult control and experimental mice were implanted with indwelling unilateral or bilateral optic fibres (200µm diameter, pitch 0.9mm, 0.37 NA; Doric Lenses) at the time of AAV injection. Fibres were implanted 1.1mm posterior to Bregma, centered around the sagittal sinus, 5.7mm deep. Mice were then habituated for 2–5 weeks prior to undergoing the optogenetic activation and bleeding protocol.

For Chr2 optogenetic studies, intact diestrous female and male mice were connected to the laser and 15-min later given 5-ms pulses of blue light (473nm, 5mW; DPSS laser, Ikecool, USA) for one min at a frequency of 10 Hz, repeated three times over 30-min intervals. Serial 3µL blood samples were collected from the tail tip at 12, 6, 0, 3, 6, 9, 12, 18, 24 and 30 min, where 0 was the start of the laser stimulation. Statistical analysis of evoked LH values was undertaken using one-way repeated measures ANOVA with Dunnett's *post hoc* test, comparing values to the 6min LH level.

For inhibitory optogenetic experiments, mice GDx at the time of AAV injection were attached to the laser patch cord in their home cages, left to acclimate for 15 min, and 5-min blood sampling (3µL per sample) undertaken for 180 min as described above. The laser was turned on for a 30-min period after 90 min of baseline sampling. Illumination with constant green light from a DPSS laser (532nm, ~5mW per fibre; Ikecool) was controlled by a Master9 stimulator. As a control in ArchT experiments, 4 mice underwent a second bleeding experiment, a least 2 weeks after the first, in which they received blue light illumination (472nm). Analysis of the frequency and amplitude of pulses was determined for the 0–85min baseline period, for the 30min laser illumination period, and for the two 30min periods thereafter. Taking insight from the fiber photometry results, an LH pulse in GDx mice was defined as a peak level of LH >10% above the preceding value. For each detected pulse, the amplitude was determined by subtracting the highest LH value from the basal value immediately prior to the onset of the pulse. Data were compared using one-way repeated measures ANOVA with *post hoc* Dunnett's tests comparing the baseline with and post illumination time periods.

Immunohistochemistry:

Immunofluorescence was undertaken as reported previously (24). Coronal brain sections (30µm thick) covering the full length of the ARN were used to determine the locations of optic fibers and/or processed for dual-label analysis. Antisera against GFP (1:5000; Aves Labs Inc.) was used to detect GCaMP6. Dual fluorescence images were captured on a NikonA1+ inverted confocal microscope. Two sections at each of the rostral, middle and caudal levels of the ARN (Suppl. Fig.5) were analyzed in each mouse by counting the total number of cells that expressed τ GFP or tdTomato, reporting kisspeptin expression, and either GFP (GCaMP6), tdTomato (ArchT), or eYFP (eNpHR3.0).

Acknowledgements

SYH and JC contributed equally to these studies with the order of names determined by coin toss. These studies were supported by the New Zealand Health Research Council, New Zealand Royal Society Marsden Fund and the BBSRC (BB/K003178/1). The authors thank Mr. Leo Van Rens, EMTech, University of Otago, for technical assistance.

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Figure Legends

Figure 1. ARN^{KISS} neurons exhibit periodic synchronizations in intracellular calcium concentrations tightly correlated with pulsatile LH secretion in gonadectomized male mice.

A. KISS1-Cre mice injected with Cre-dependent GCaMP6s AAV into the ARN (green) and connected to a fiber photometry system with regular blood sampling from the tail tip. Alternating 490-nm (Ca^{2+} -dependent) and 405-nm (Ca^{2+} -independent) wavelength light passes down the photometry fiber. **B.** Simultaneous GCaMP6s calcium (top) and cell-attached (bottom) recordings from an ARN^{KISS} neuron. numbers = spikes per burst. **C.** Fluorescence emission traces (490-nm calcium=blue, 405-nm background = purple) recorded from ARN^{KISS} neurons of a gonadectomized male mouse with the subtracted signal (490–405) shown below. **D.** Continuous recording of two calcium events from a gonadectomized mouse. **E.** Average fluorescence waveform of calcium events (16 events, 7 mice) normalized to beginning of each event. **F.** Simultaneous brain slice recording of GCaMP6s fluorescence (top) and action potential firing (bottom) from a GCaMP6s/hM3Dq-expressing ARN^{KISS} neuron showing the response to CNO applied in the bath. **G.** Representative trace showing effect of i.p. CNO (arrow) on GCaMP6 calcium signal from ARN^{KISS} neurons in a gonadectomized mouse. **H & I.** Traces showing the near perfect correlation of ARN^{KISS} neuron GCaMP6 calcium events (black) with pulses of LH secretion (red) in a gonadectomized male mouse. **I.** Shows the mouse with the slowest calcium events.

Figure 2. Brief 1-min channelrhodopsin activation of ARN^{KISS} neurons evokes pulsatile LH secretion in freely behaving intact male and female mice.

A. Schematic of experiment. **B.** Repeated 1-min blue light (473nm) activations of ARN^{KISS} neurons evoke LH pulses in an intact male mouse. **C.** Mean ($\pm$ SEM) LH levels (red symbols indicate significantly elevated LH levels; repeated measures ANOVA with Dunnett's *post hoc* tests; $n=5$). **D.** Repeated 1-min blue light (473nm) activations of ARN^{KISS} neurons evoke LH pulses in an intact female mouse. **E.** Mean ($\pm$ SEM) LH levels (red symbols indicate significantly elevated LH levels; repeated measures ANOVA with Dunnett's *post hoc* tests; $n=6$). **F.** Overlayed profiles of endogenous (red) and ChR2-activated (blue) LH pulses in intact male mice. Pulses normalized for fold-increase over baseline; endogenous LH pulses ($n=6$) sampled every 10 min and ChR2-activated LH pulses ($n=15$) every 6 min.

Figure 3. Archaelhodopsin inhibition of ARN^{KISS} neurons suppresses pulsatile LH secretion in gonadectomized female mice.

A. KISS1-Cre mice injected with Cre-dependent ArchT tdTomato AAV into the ARN (red) and implanted with bilateral optic fibers in the ARN. **B.** Dual fluorescence image of ARN^{KISS} neurons expressing ArchT (red) and kisspeptin (GFP reporter). **C.** Action potential firing in an ArchT-expressing ARN^{KISS} neuron in the presence of 10nM NKB in a brain slice. The neuron responds to green light illumination (shading) with a decrease in firing followed by a return to control levels. Top trace, action potential firing. Bottom trace, rate meter for the same cell. **D.** Mean ($\pm$ SEM) normalized firing rate of ARN^{KISS} neurons ($n=11$) responding to green light. * $p<0.05$ compared with baseline (Friedman test). **E-H.** Pulsatile LH secretion in ArchT-KISS1-Cre (E-G) and wild type (H) gonadectomized female mice. Green light illumination (30 min) indicated by green shading, LH pulses by asterisks. Trace in **G.** shows the same

mouse in **F**, but illuminated with blue light. **I**, Mean ($\pm$ SEM) LH levels in KISS1-Cre mice (n=6) showing the suppression of LH secretion during green light illumination and subsequent slow recovery. **J & K**, Mean ($\pm$ SEM) LH pulse amplitude and pulse frequency prior to (0-85 min), during (90-115 min; green shading) and subsequent (120-145 and 150-175 min) to green laser illumination. * $p < 0.05$, ** $p < 0.01$ versus 0-85 min; ANOVA with Dunnett's *post hoc* tests, n=6. **L-N**, Basal LH levels, LH pulse amplitude and frequency in control mice including blue light-illuminated KISS1-Cre and wild-type AAV-injected mice (n=8).

Figure 4. Halorhodopsin inhibition and activation of ARN^{KISS} neurons suppresses and then initiates on-going pulsatile LH secretion in gonadectomized female mice.

A, KISS1-Cre mice injected with Cre-dependent eNpHR3.0-eYFP AAV into the ARN (green) and implanted with bilateral optic fibers in the ARN. **B**, Dual fluorescence image of ARN^{KISS} neurons expressing eNpHR3.0 (green) and kisspeptin (tdTomato reporter). **C**, Action potential firing in an eNpHR3.0-expressing ARN^{KISS} neuron in the presence of 20nM NKB. The neuron responds to green light with an abrupt decrease in firing followed by a sharp rebound activation immediately after the light is stopped. Top trace, action potential firing. Bottom trace, rate meter. **D**, Mean ($\pm$ SEM) normalized firing rate of ARN^{KISS} neurons (n=6) responding to green light. * $p < 0.05$ compared with baseline (Friedman test). **E-H**, Pulsatile LH secretion in eNpHR3.0-KISS1-Cre (E,F) and wild-type (G,H) gonadectomized female mice. Green light illumination indicated by green shading and LH pulses by asterisks. **I**, Mean ($\pm$ SEM) LH levels in KISS1-Cre mice (n=8) showing the suppression of LH secretion during green light illumination and subsequent rebound of LH and re-setting of pulse generator to evoke a subsequent LH pulse in all mice (arrow). **J & K**, Mean ($\pm$ SEM) LH pulse amplitude and pulse frequency prior to (0-85 min), during (90-115 min; green shading) and subsequent (120-145 and 150-175 min) to green laser illumination. * $p < 0.05$, ** $p < 0.01$ versus 0-85 min; ANOVA with *post hoc* Dunnett's test, n=8. **L-N**, Basal LH levels, LH pulse amplitude and frequency in control AAV-injected wild-type mice (n=7).